

A Ditlevsen–Karppinen transition-density competitor for the Dacca cholera benchmark

Computational note

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1 Question and evaluation target

Reviewer 1 proposed constructing tractable Gaussian transitions using the strong-order-1.5 approximation of Ditlevsen and Samson (DS), then using those transitions in SAEM, Fisher-identity gradient estimation, PGAS, or a particle Gibbs method stabilized by Karppinen, Singh, and Vihola (KSV). The statistical criterion in the DMOP paper is not the complete-path pseudo-likelihood. It is the log likelihood of the Euler-discretized POMP, independently evaluated by a bootstrap particle filter. Accordingly, a candidate method is successful only if its estimated parameter vector has a competitive Euler-model log likelihood.

The paper used 100 searches from the same wide box, compared methods under similar elapsed time, evaluated selected parameter vectors with 5,000 particles and 36 replications, and emphasized both the right tail and median of the final log likelihoods. Its best comparable-effort result is -3744.17 for IFAD-0.97; extended IFAD-0.97 reaches -3744.19 with median -3744.22 .

2 Dacca state representation

For the benchmark, $c = \alpha = 1$, $\rho = Y_0 = 0$, and $\delta = 0.02$ are fixed. Thus the inapparent-infection coordinate remains zero. In population-fraction coordinates, the active state is

$$X = (S, I, R^1, R^2, R^3, M)^\top \in \mathbb{R}^6,$$

where M accumulates disease deaths within a month and is reset after each measurement. With drift $b_\theta(X, t)$, the single diffusion vector field is

$$g_\theta(X, t) = \sigma SI(-1, 1, 0, 0, 0, 0)^\top, \quad dX_t = b_\theta(X_t, t)dt + g_\theta(X_t, t)dB_t.$$

The observation model is exactly the manuscript model, $Y_n \mid X_{t_n} \sim \mathcal{N}(P_{t_n} M_{t_n}, \tau^2 P_{t_n}^2 M_{t_n}^2)$.

3 Diffusion rank, transition rank, and bracket depth

It is essential not to confuse the two ranks. Hypoellipticity means that the *instantaneous diffusion matrix is not full rank* while the finite-time transition nevertheless has a smooth density. DS assume $V \in \mathbb{R}$, $U \in \mathbb{R}^p$, and

$$C(v, u) = \begin{bmatrix} 0_{1 \times p} \\ \Gamma(v, u) \end{bmatrix}, \quad \Gamma(v, u) = \text{diag}\{\sigma_1(v, u), \dots, \sigma_p(v, u)\}, \quad \sigma_j > 0.$$

Thus C has rank $p < p + 1$ by construction. The directly forced rough block already supplies p directions and the first drift bracket supplies the one missing smooth direction. The order-1.5 term is exactly what gives their leading approximate covariance a nonzero variance of order Δ^3 in that smooth coordinate for sufficiently small Δ .

Equivalently, the DS instantaneous diffusion has rank $d - 1$ in a d -state system. Dacca has $d = 6$ active coordinates but diffusion rank one, hence codimension five rather than one. Diffusion rank is invariant under an invertible change of state coordinates, so state dependence of the coefficient σSI does not put Dacca into the DS model class.

Dacca is potentially hypoelliptic too, but it is not in this depth-one form. Its active state has dimension six and its instantaneous diffusion has rank one away from $SI = 0$ and rank zero on that boundary. At the manuscript parameter and state point, the normalized columns formed with the Stratonovich drift $b^S = b - \frac{1}{2}(Dg)g$,

$$g, [b^S, g], [b^S, [b^S, g]], \dots$$

gain numerical ranks 1, 2, 3, 4, 5, 6 only through bracket depth five. This is evidence along one repeated drift-bracket chain that the true finite-time SDE transition can be full-dimensional; it is neither a proof of global hypoellipticity nor an argument that the diffusion should be full rank.

The difficulty is instead that a literal first-bracket generalization of the DS leading covariance does not contain all those directions. Let $J_b = \partial b_\theta / \partial x$. Locally freezing the state-dependent diffusion gives

$$Q_h^{(1.5)}(x) = hgg^\top + \frac{h^2}{2}\{(J_b g)g^\top + g(J_b g)^\top\} + \frac{h^3}{3}(J_b g)(J_b g)^\top \quad (1)$$

$$= A_2 H_2 A_2^\top, \quad (2)$$

$$A_2 = \begin{bmatrix} h^{1/2}g & h^{3/2}J_b g \end{bmatrix}, \quad (H_2)_{ij} = \frac{1}{i+j+1}, \quad i, j \in \{0, 1\}. \quad (3)$$

Therefore

$$\text{rank}\{Q_h^{(1.5)}(x)\} \leq 2 < 6.$$

This conclusion is not invalidated by the state dependence highlighted by the reviewer. Here $g(x) = q(x)e$ with the constant direction $e = (-1, 1, 0, 0, 0, 0)^\top$, so $(Dg)w = e\{(Dq)w\}$ is collinear with g for every w . The Milstein and other diffusion-derivative terms in the full strong-order-1.5 expansion therefore remain on the directly forced line; the first drift-propagation term adds only the $J_b g$ direction. The paper gives its nondegenerate Gaussian construction for the $1 + p$ structure with p directly forced directions, not a six-state/one-driver, depth-five analogue. This rank-two statement concerns the approximate *finite-step covariance*, not the rank-one instantaneous diffusion. It shows that the DS nondegeneracy argument for their $1 + p$ depth-one model cannot simply be copied to this longer-chain system. Further stochastic Taylor/bracket terms, or a block construction that accumulates them, are required. The implementation also uses the DS second-order mean

$$m_{\theta,h}(x) = x + hb_\theta(x) + \frac{h^2}{2}\mathcal{L}_\theta b_\theta(x),$$

where $\mathcal{L}_\theta$ is the Itô generator.

Adding a nugget makes numerical evaluation possible but changes the statistical model, with the nugget dominating directions that the first-bracket expansion leaves exactly deterministic. Results labeled “DS-local+nugget QML” disclose this change and are not presented as a literal execution of DS Algorithm 2.

4 Which time scale is being compared

One Euler step per measurement interval is indeed the measurement time scale. The important distinction is between transition *schemes* on that same grid. With r substeps per month,

$r = 1$: one monthly DS transition, $r = 20$: twenty DS transitions of length 1/20 month.

DS demonstrate inference with their discretization points at the observation times. Introducing $r - 1$ intermediate latent states per month is an extension. In every case below, goodness is assessed against the *same* numerical model used by the DMOP manuscript: the Euler POMP with 20 substeps per month. A one-step Euler likelihood is therefore neither the target nor evidence about the one-step DS fit.

Grid equality also says nothing about numerical stability. The published Dacca point has a three-stage immunity rate $3\epsilon = 57.3$ per year. Applied to a decay mode with rate λ , the DS second-order conditional mean has stability polynomial $R(z) = 1 - z + z^2/2$, $z = h\lambda$, whose real-axis absolute-stability interval is $0 \leq z \leq 2$. The monthly, half-monthly, and five-substep values of $h(3\epsilon)$ are respectively 4.78, 2.39, and 0.96. Thus failure at $r = 1, 2$ and viability beginning at $r = 5$ are the expected coarse-step behavior of this mean, not evidence that the measurement and one-step grids were different.

5 Higher-bracket completion as a diagnostic

To test the strongest nearby density-based idea, define the local controllability expansion

$$A_q(x, h) = \left[h^{1/2}g, \frac{h^{3/2}}{1!}J_b g, \dots, \frac{h^{q-1/2}}{(q-1)!}J_b^{q-1}g \right], \quad (4)$$

$$(H_q)_{ij} = \frac{1}{i+j+1}, \quad Q_h^{(q)}(x) = A_q H_q A_q^T. \quad (5)$$

This is the covariance obtained by integrating the Taylor expansion of $\exp(J_b s)g$ over a step. For $q = 2$ it reproduces the DS leading covariance; for $q = 6$ the unscaled Dacca local controllability matrix is generically full rank. The mean remains the DS second-order generator expansion. This “DS–Krylov” completion is a new local-linear Gaussian approximation, not a claim about the scope of the DS paper. At fine h , its smallest directions scale as high odd powers of h , so the transition becomes extremely ill-conditioned even when it is algebraically full rank. It is implemented as a rank diagnostic and optional sensitivity analysis. The primary executable competitor instead uses the literal two-column DS leading covariance plus a reported relative numerical nugget; neither choice is presented as an exact application of DS to Dacca.

The earlier extended-Kalman quasi-likelihood optimizer has been withdrawn and is not used as a DS result. The executable competitor now uses a bootstrap particle filter at every parameter iteration, draws one path from the forward genealogy, evaluates the complete Gaussian pseudo-likelihood on that path, and takes a projected stochastic Fisher-score step. The score average uses the DS gain schedule. Because the Dacca complete pseudo-likelihood has not been shown to have DS’s curved exponential-family form, there is no SAEM M-step in the executable method: this is stochastic gradient ascent, not DS Algorithm 2. Every candidate is judged afterward by the original Euler-20 particle filter rather than by its training pseudo-likelihood.

More precisely, write $\tilde{p}_{\Delta, \theta}$ for the regularized DS Gaussian pseudo-model with latent path X . Fisher’s identity gives

$$\nabla_{\theta} \tilde{\ell}_{\Delta}(\theta; y) = \mathbb{E}_{\theta}[\nabla_{\theta} \log \tilde{p}_{\Delta, \theta}(X, y) \mid y].$$

At iteration m , forward SMC approximates $\tilde{p}_{\Delta, \theta_{m-1}}(X \mid y)$ and one complete genealogy $X^{(m)}$ is drawn from its terminal weighted empirical measure. Automatic differentiation is then applied to $\log \tilde{p}_{\Delta, \theta}(X^{(m)}, y)$ while holding $X^{(m)}$ fixed. Thus

$$\hat{z}_m = \nabla_{\theta} \log \tilde{p}_{\Delta, \theta}(X^{(m)}, y) \Big|_{\theta=\theta_{m-1}}$$

is a one-path SMC approximation of the pseudo-model observed score. It is not an estimate of the original Dacca SDE/Euler score: the latter transition density is unavailable, and the first-bracket Dacca covariance is singular until the disclosed nugget is added. No derivative is taken through particle simulation, resampling, or ancestor selection. Updating $s_m = (1 - a_m)s_{m-1} + a_m \hat{z}_m$ and stepping along s_m is a Robbins–Monro score method. In contrast, proper SAEM updates a parameter-independent sufficient statistic $S(X^{(m)}, y)$ and maximizes its surrogate exactly (or with a justified generalized-EM step).

6 Why KSV is relevant, and why it is not a direct patch

At $r = 1$ there are no artificial within-month latent points, so KSV does not solve the monthly Taylor-approximation or rank-depth issues. It can become relevant when DS is extended to $r > 1$. In ordinary particle backward sampling, the ancestor weight contains $p_h(x_{t+h} \mid x_t)$. As h decreases, this density concentrates and almost all backward weight is assigned to the existing ancestor. Increasing the number of Euler steps between monthly observations therefore degrades path mixing. KSV replace unit-step backward sampling by bridge updates spanning a fixed physical-time block. Their method directly targets this failure mode.

There is an important algorithmic distinction. DS Algorithm 2 draws a fresh path from an ordinary forward SMC approximation at every SAEM iteration; it does not require a conditional-particle-filter Markov kernel. KSV specifically stabilize the backward update of a CPF/particle-Gibbs kernel. Consequently KSV is not required merely to run the DS-style forward-SMC competitor with twenty substeps. It would be relevant if that independent imputation were replaced by particle Gibbs or another CPF path update.

KSV Assumption 7 requires evaluation of multi-step densities $M_{u|\ell}(x_u \mid x_{\ell})$ and simulation from bridge conditionals $M_k(x_k \mid x_{k-1}, x_u)$. The nonlinear, state-dependent DS local transition does not supply these objects. A valid combination must introduce a tractable time-varying linear-Gaussian proposal M and put the density correction into the Feynman–Kac potentials. KSV also does not repair the rank-two DS transition, and Dacca’s complete likelihood is not the curved exponential family required for DS’s closed-form SAEM M-step. One must either derive a valid generalized-EM surrogate or use a separately labeled score method.

Algorithm 1 is the minimal coherent combination of the two papers for Dacca. It is materially more than “put KSV inside DS”: it requires a higher-order rank completion, an auxiliary Gaussian bridge model, exact importance correction for that proposal, a forward conditional-particle pool, and a numerical M-step.

The code implements a narrower fixed-lower/fixed-upper blocked-interior conditional SMC kernel derived from the bridge construction in KSV Algorithm 8. Its auxiliary bridge conditionals and p^K/M corrections are internally consistent, but it cannot reconnect the upper endpoint to a different lower-boundary forward particle. It is therefore neither full Algorithm 8 nor CPF–BBS, and its fraction of changed interiors is not KSV’s probability of lower-boundary update (PLU). Short Dacca checks found mixed effects on parameter progress and no repair of the one-ancestor forward genealogy. It is excluded from the headline baseline unless a full forward CPF–BBS implementation passes a separate invariance and mixing validation.

Input: observations $y_{1:N}$; initial θ_0 ; Euler step h ; particles J ; SA gains a_m ; physical-time block endpoints $\mathcal{T}$.

1. Construct the higher-bracket completed transition

$$p_{\theta,h}^K(x_k | x_{k-1}) = \mathcal{N}\left(m_{\theta,h}(x_{k-1}), Q_{\theta,h}^{(6)}(x_{k-1})\right).$$

2. At the current reference path, construct a globally time-varying linear-Gaussian proposal $M_{\theta,1:T}$ whose multi-step marginals and bridge conditionals are available by Kalman recursions.
3. Define correction potentials

$$G_{\theta,k}(x_{k-1}, x_k) = \frac{p_{\theta,h}^K(x_k | x_{k-1})}{M_{\theta,k}(x_k | x_{k-1})} \times g_{\theta}(y_n | x_k)^{\mathbf{1}_{\{k=t_n\}}}.$$

4. For $m = 1, 2, \dots$:

- (a) Run a conditional particle filter for $(M_{\theta_{m-1}}, G_{\theta_{m-1}})$.
- (b) Apply KSV bridge backward sampling over $\mathcal{T}$, proceeding from the final block to the first and using the Gaussian bridge conditionals of M , to obtain $X_{0:T}^{(m)}$.
- (c) Compute the complete-data score $z_m = \nabla_{\theta} \{\log p_{\theta}(X_0^{(m)}) + \sum_k \log p_{\theta,h}^K(X_k^{(m)} | X_{k-1}^{(m)}) + \sum_n \log g_{\theta}(y_n | X_{t_n}^{(m)})\}$.
- (d) Update the stochastic approximation $s_m = (1 - a_m)s_{m-1} + a_m z_m$.
- (e) Take a constrained stochastic-score parameter step, e.g. $\theta_m = \Pi_{\Theta}\{\theta_{m-1} + \eta_m \text{Adam}(s_m)\}$.

Output: $\hat{\theta}$ and an independent Euler-particle-filter estimate of $\ell_{\text{Euler}}(\hat{\theta})$.

Algorithm 1: Full DS–Krylov–KSV stochastic-approximation score ascent (design, not the executed baseline)

The executed numerical score method is a Fisher-identity Robbins–Monro extension, rather than the closed-form curved-exponential-family SAEM used by DS. Neither it nor the blocked-interior diagnostic is used to claim that full Algorithm 1 has been run.

References

- [1] S. Ditlevsen and A. Samson (2019). Hypocoelliptic diffusions: filtering and inference from complete and partial observations. *Journal of the Royal Statistical Society, Series B*, 81, 361–384. arXiv:1707.04235.
- [2] S. Karppinen, S. S. Singh, and M. Vihola (2024). Conditional particle filters with bridge backward sampling. *Journal of Computational and Graphical Statistics*, 33, 363–378.